Models

<<abstract>>
Campus Asset

```
-purchasePrice:decimal
-status: AssetStatus
+AssetTag: string
+Name: string
+PurchaseDate: DateTime

+CampusAsset(tag: string, name: string,
purchaseDate: DateTime, price: decimal)
+CampusAsset()
+MarkInRepair()
+MarkAvailable()
+Retire()
+SetCheckedOut()
+SetReturned()
#GetMaintenanceStatus():string
#CalculateDepreciation(): decimal
+ToString(): string
```

Asset Status

```
+Available:enum
+CheckedOut: enum
+InRepair:enum
+Retired:enum
```

Software License

```
+ Vendor: string
+ SeatCount: int
+ SubscriptionEndDate: DateTime

+ SoftwareLicense(tag: string, name: string,
  purchaseDate: DateTime, price: decimal,
  vendor: string, seats: int,
  endDate: DateTime)
+ GetMaintenanceStatus(): string
+ CalculateDepreciation(): decimal
```

Networking Hardware

```
+DeviceModel:string
+PortCount: string
+IpAddress: string
+MacAddress: string

+NetworkingHardware(tag: string, name: string,
  purchaseDate: DateTime, price: decimal,
  model: string, ports: int)
+ GetMaintenanceStatus(): string
+ CalculateDepreciation(): decimal
+ HasNetworkIdentity(): bool
```

Workstation

```
+ Cpu: string
+ RamGb: int
+ LabRoom: string
+ BorrowerId: string
+ CheckoutDate: DateTime?
+ IpAddress: string
+ MacAddress: string

+ IsCheckedOut: bool
+ Workstation(tag: string, name: string,
  purchaseDate: DateTime, price: decimal,
  cpu: string, ram: int, room: string)
+ GetMaintenanceStatus(): string {override}
+ CalculateDepreciation(): decimal {override}
+ CheckOut(borrowerId: string): void
+ Return(): void
+ HasNetworkIdentity(): bool
```

Peripheral

```
+ DeviceType: string
+ BorrowerId: string
+ CheckoutDate: DateTime?

+ IsCheckedOut: bool { get }
+ Peripheral(tag: string, name: string,
  purchaseDate: DateTime, price: decimal, type:
string)
+ GetMaintenanceStatus(): string {override}
+ CalculateDepreciation(): decimal {override}
+ CheckOut(borrowerId: string): void
```

Asset Response

```
+AssetType:string
+AssetTag:string
+Name:string
+PurchaseData: DateTime
+PurchasePrice:decimal
+Cpu:string
+RamGb:int
+LabRoom:string
+DeviceModel: string
+PortCount: int
+DeviceType:string
+Vendor: string
+SeatCount:int
+SubscriptionEndDate: DateTime
+IpAddress: stirng
+MacAddress: string
```

Assest Request

```
+AssetType:string
+AssetTag:string
+Name:string
+PurchaseData: DateTime
+PurchasePrice:decimal
+Cpu:string
+RamGb:int
+LabRoom:string
+DeviceModel: string
+PortCount: int
+DeviceType:string
+Vendor: string
+SeatCount:int
+SubscriptionEndDate: DateTime
+IpAddress: stirng
+MacAddress: string
```

Interfaces

Iborrowable

```
+AssetTag: string
+BorrowerID: string
+IsCheckedOut: bool

+CheckOut(string borrowerID)
+ return()
```

INetiworkable

```
+IpAddress: string
+MacAddress: string

+ HasNetworkIdentity()
```

Controllers

Asset Control

```
- service: AssetService

+ AssetController(service: AssetService)
+ AddAsset(asset: CampusAsset): string
+ GetAsset(tag: string): CampusAsset
+ ListAll(): List<CampusAsset>
+ DeleteAsset(tag: string): string
+ SendToRepair(tag: string): string
+ MarkRepaired(tag: string): string
+ CheckOut(tag: string, borrowerId: string): string
+ Return(tag: string): string
+ GetCheckedOut(): List<IBorrowable>
+ Search(text: string): List<CampusAsset>
+ GetByType(typeName: string): List<CampusAsset>
+ GetInRepair(): List<CampusAsset>
+ GetNetworkProblems(): List<INetworkable>
+ GetTotalValue(): decimal
```